Management Science

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MANAGEMENT SCIENCE
Vol. 71, No. 6, June 2025, pp. 4533–4551
ISSN 0025-1909 (print), ISSN 1526-5501 (online)

How Does a Health Insurance Program Covering 500 Million Poor Impact Credit Market Outcomes?

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Received: April 4, 2023
Revised: October 4, 2023; January 16, 2024
Accepted: February 12, 2024
Published Online in Articles in Advance: September 24, 2024

https://doi.org/10.1287/mnsc.2023.01039

Abstract. I study the impact of the world's largest publicly funded health insurance plan for the poor on credit markets. India launched a health insurance plan that covered 500 million beneficiaries. The fact that opposition-ruled states did not implement the program for political reasons allows me to compare border districts of implemented and nonimplemented states within a difference-in-differences framework. I find that loan delinquency reduced significantly in implemented districts. Increased liquidity for loan repayment seems to be the mechanism at work. I rule out the federal-level ruling party's influence as an explanation.

History: Accepted by Kay Giesecke, finance.
Supplemental Material: The online appendix and data files are available at https://doi.org/10.1287/mnsc.2023.01039.

Keywords: health insurance • household finance • banking • access to finance

1. Introduction

Large medical bills could disrupt the economic lives of the poor, especially the uninsured. Therefore, the literature on health insurance has emphasized the importance of considering the direct financial gains from public health insurance programs in addition to any indirect benefits from an effect on health (Finkelstein and McKnight 2008). In this context, I study the world's largest publicly funded health insurance program implemented by the Indian government (Pradhan Mantri Jan Arogya Yojana (PMJAY)) and ask whether it improved loan performance.

The expected impact is not clear. As pointed out by Cole et al. (2013), poor people face several nonprice barriers, such as lack of trust, liquidity constraints, and limited salience in adopting insurance. Further, health infrastructure is significantly underdeveloped. Therefore, an increase in demand for healthcare because of PMJAY could lead to failed or delayed treatments and cause further aversion to health insurance and formal medical facilities (Das et al. 2016). Thus, there are reasons to believe that PMJAY would have no impact.

However, the implementation of a publicly funded health insurance program could help overcome the barriers described above. In that case, it may reduce the need for precautionary savings, reduce out-of-pocket expenses on health, and consequently, improve liquidity for loan repayments. To the extent that health insurance helps the poor access medical treatment promptly, it may reduce economic disruptions and lead to better loan performance (Coile 2004, Wagstaff 2007). Access to preventive healthcare enabled by health insurance may also reduce the incidence of illness, enhance productivity, and lead to lower loan defaults.

The Indian government's universal health insurance program, PMJAY, provides the economic setting for this study. The first leg of the program covered 107.4 million low-income families consisting of nearly 500 million people. The families were identified based on the pre-existing socioeconomic caste survey. The families identified as "deprived" by the survey were automatically enrolled into the program. The program offered insurance coverage of Indian rupees (INRs) 500,000 per household per year.

PMJAY is the most comprehensive health insurance program introduced by the Government of India. The insurance plan covers the cost of treatment, medicines, testing, and incidental expenses incurred by the patient and one attendant. The insured can access any empaneled health facility. The sheer size of the program and the consequent implementation-related issues make it worthy of study. I focus on the program's impact on credit markets.

I obtain data relating to loans from one of the largest credit bureaus in India. The bureau provided data at a district-lender type-area type-loan cohort quarter-loan repayment quarter level. Within a district, areas are divided into urban, rural, and semiurban categories. Bank types include government banks, private banks, nonbanks, rural banks, small finance banks, microfinance institutions, etc. Loan cohort refers to the calendar quarter in which a loan is issued. I obtain data relating to monthly household income and expenditure from the Centre for Monitoring Indian Economy (CMIE). I use microfinance loans for my primary analysis. As I explain in detail in Section 4, borrowers of microfinance loans are likely to be representative of PMJAY beneficiaries based on their income levels and the level of asset holdings. I conduct robustness tests using small agricultural loans. The measure of loan outcome is the classification of a loan as a nonperforming asset (NPA).

I use the fact that the states ruled by regional opposition parties did not implement PMJAY, plausibly because of political reasons, for identification. I compare border districts of nonimplemented states with contiguous border districts of implemented states within a difference-in-differences (diff-in-diff) framework. The expected similarity between neighboring districts helps address concerns relating to the existence of correlated time-varying shocks.

I start the empirical analysis with a first-stage test. Using two national family health surveys, I find that the health insurance coverage in implemented border districts went up by 152% more after the program implementation than their contiguous districts belonging to the nonimplemented states. My main diff-in-diff tests using microfinance loans show that the NPA rate is 3.4 percentage points lower in implemented districts when compared with neighboring nonimplemented districts. Given the average NPA rate of 9.83% before PMJAY, the difference represents a 35% change, which is economically meaningful. The test suggested by Rambachan and Roth (2023) helps address the concern that the results documented above are a mechanical continuation of pre-existing trends. I find similar results using the sample of small agricultural loans.

To my knowledge, no other interventions moved with PMJAY and impacted the implemented and nonimplemented regions differently, except a cash transfer program for farmers. The cash transfer was not implemented by one of the PMJAY nonimplementing states. All other states implemented it. My results hold when I (i) exclude that state and its neighbors from the analysis and (ii) compare the loan performance of that state with the loan performance of a neighboring state that did not implement PMJAY but implemented the cash transfer program.

Next, I address the concern that the nonimplemented states rejected PMJAY because they already had better state-level health insurance schemes in several ways. First, I compare the loan performance of implemented and nonimplemented states within a sample of states having a local health insurance program. I find that the implemented states outperform even in this sample. Second, I compare features of state-level insurance schemes and find them less comprehensive compared with PMJAY. Third, I do not find a significant difference between the control and treatment regions in terms of their budget allocations to the health sector. Fourth, health outcomes, health insurance coverage, and macroeconomic conditions before implementation are broadly similar between the two groups of states. Superior health schemes, if they existed, should have led to better health outcomes and insurance coverage in nonimplemented states. Finally, given that PMJAY gives flexibility to states to both retain their own schemes and also top up PMJAY with extra benefits, purely from an economic point of view, it is not rational for states to deny an additional 60% funding of their health budget. Therefore, it is more likely that the reasons for nonimplementation are political and not the existence of a state-level health insurance scheme.

A reader may worry that the documented change in credit market outcomes is because of the ruling party at the federal level favoring states ruled by it or its allies. I call it the "ruling party effect." I conduct several tests to rule out the above explanation. First, using the fact that the main national-level opposition party-ruled states implemented the program, I compare such states with the nonimplementing states in terms of the performance of microfinance loans. I find that even in this subsample, the PMJAY-implemented regions outperform. Second, I compare the national opposition party-ruled border regions and neighboring regions ruled by the federal ruling party. Because the treatment and control regions do not differ concerning PMJAY treatment, I expect to find no difference in the loan performance of PMJAY-impacted loans. The results are in line. Finally, the placebo tests with ineligible borrowers also help me rule out the ruling party effect.

In the last part of the paper, I attempt to identify the mechanism. A publicly funded health insurance scheme can improve loan performance by increasing the liquidity available for loan repayment. Using monthly household-level consumer survey data, I find that PMJAY leads to a reduction in precautionary savings. The results suggest that in the absence of health insurance, the utility of having funds for health emergencies dominated the costs of loan default. PMJAY reduced the need to make such a trade-off. Further, I find a decline in out-of-pocket expenditure on health. Finally, I also find a reduction in borrowings from moneylenders for health emergencies. Thus, PMJAY reduced the need to borrow expensive loans for health emergencies and increased the funds available for repaying other loans. Given the above evidence, it is reasonable to conclude that PMJAY increased liquidity for loan repayment.

The paper contributes to the large and growing literature on the impact of health insurance (Gruber and Simon 2008, Bharadwaj et al. 2013, Baicker et al. 2014, Garthwaite et al. 2014, Mahoney 2015, Dobkin et al. 2018, Goodman-Bacon 2018, Abaluck et al. 2021, Goldin et al. 2021, Miller et al. 2021a). Within this literature, my work is related to studies that examine the impact of publicly funded health insurance on financial well-being. Gallagher et al. (2019) show that a federal subsidy on marketplace insurance leads to a decline in home mortgage and rent repayment delinquencies. Goldsmith-Pinkham et al. (2020) examine the impact of health insurance provided by the Medicaid program. They find a significant decline in debt in collections but not in other measures of financial health, most notably loan delinquency. Other studies show that improved access to health insurance is associated with a reduction in out-of-pocket health expenses and in new debt to pay for healthcare (Finkelstein et al. 2012, Shigeoka 2014, Levine et al. 2016, Allen et al. 2017, Hu et al. 2018, Miller et al. 2021b).

This paper adds to the knowledge of the impact of health insurance on financial well-being in four different ways. First, given the type of beneficiaries I study and the health infrastructure situation in emerging economies, a strong null hypothesis would be of no effect. As I describe in detail in Section 2.1, the beneficiaries (i) do not have access to private health insurance, (ii) face significant price and nonprice barriers in taking up insurance, and (iii) have to depend on an underdeveloped health infrastructure. The types of beneficiaries that I study likely represent the majority of the world's poor not having access to health insurance. This differs from the setting studied by the extant literature that links health insurance and financial well-being. For instance, the insurance subsidy that Gallagher et al. (2019) study is extended to people slightly above the poverty line in the United States. Offering subsidies increases insurance coverage as the target segment is not otherwise averse to the idea of insurance. Further, the beneficiaries file tax returns and have records of their financial transactions, which allows private insurance companies to assess risk and offer insurance policies. As the adoption rates shown in the paper suggest, although price barriers matter, the nonprice barriers do not seem very important in their setting. Thus, I show that the provision of publicly funded health insurance positively impacts the financial outcomes for much broader and more diverse beneficiaries than studied by the extant work.

Second, the program's design, scale, and implementation make it worthy of study. Given the absence of a private insurance market dealing with the beneficiaries, the program's design had to be different than the publicly funded insurance programs in developed countries. The government had to act as a quasi-insurance company by actuarially estimating the claim amount and setting aside the same from their budget. They could not buy or subsidize the purchase of insurance policies. Further, the program covered nearly 500 million people. The beneficiaries were identified through a survey and not based on financial or other records. Studying the likely outcomes of such a program is likely to be useful to policymakers in designing health insurance programs for most of the world's poor.

Third, the dominant mechanism that I identify differs from all of the above papers. The extant findings show that subsidies on premiums and cost-sharing reduce out-of-pocket health expenditure, leaving households with more resources to fulfill their other obligations. In contrast, my results suggest that access to health insurance can reduce loan delinquency even if the beneficiaries do not suffer health-related emergencies. This is because of a likely decline in precautionary savings, which leaves the households with more resources for loan repayment. In addition, I also find a reduction in borrowings related to health.

Finally, the financial outcome measure that I use is loan delinquency. However, all the above studies, except Gallagher et al. (2019), use the amount of unpaid bills, debt in collection, or outstanding credit as outcome measures. This distinction is important as eventual delinquency has a much higher and more direct impact on bank balance sheets. In fact, Goldsmith-Pinkham et al. (2020), who find that health insurance coverage leads to a reduction in debt in collection, do not find a significant change in loan delinquency. Gallagher et al. (2019) find an improvement in mortgage loan performance because of health insurance. Given the contrasting findings in developed market settings, documenting the positive of health insurance on loan performance in emerging economies is important.

The paper also contributes to the literature on household finance in emerging economies (Beck and Webb 2003, Gaurav et al. 2011, Badarinza et al. 2016, Anagol et al. 2017, Cole and Xiong 2017, Cole et al. 2017, Gomes et al. 2021). Badarinza et al. (2019) state that most extant research on household financial behavior in emerging economies focuses on the extensive margin. In other words, the focus is on whether households have access to formal savings and credit markets and the impact of such access (Agarwal and Mazumder 2013; Fernandes et al. 2014; Karlan et al. 2016; Dupas et al. 2018, 2019; Flory 2018). Questions relating to the type of financial instruments that the households deal with, the complementarity or substitutability of those instruments, and the differential impact of various financial instruments remain unexplored. I contribute to this literature by showing that improved access to health insurance directly impacts precautionary savings and loan performance.

2. Institutional Details

2.1. Background—Health Situation In India

The economic survey (2020–2021) released by the government describes the health situation in India. The public expenditure on health is around 1% of the gross domestic product (GDP), which is one of the lowest even among comparable countries. The out-of-pocket spending on health is close to 60%. Other vital statistics, such as infant mortality, maternal mortality, and the proportion of household expenditure on health, continue to be at elevated levels.

Most people in lower strata of income distribution work in informal sectors with no proper record of their income, occupational hazards, and lifestyle. Note that agricultural income is exempt from income tax. Given that most of the beneficiaries depend on agriculture, the tax exemption makes estimating the likely income of the beneficiaries credibly implausible. Property rights over land and other assets are not clearly demarcated (Mukherjee et al. 2018). Further, as pointed out by Cole et al. (2013), the likely group of beneficiaries faces several nonprice barriers, such as lack of trust, liquidity constraints, and limited salience, over and above price barriers. Crucially, there are no systematic and reliable records of the actual health situation of the beneficiaries and the treatments availed in the past. It is extremely hard for private insurance companies to emerge in such a setting because of the possibility of adverse selection. A publicly funded health insurance program would require the government to act as a quasi-insurance company. It is in this background that the government launched PMJAY.

2.2. The PMJAY Scheme

During the budget speech delivered on February 1, 2018, the finance minister announced an initial allocation for a health insurance scheme to cover all Indians. On September 23, 2018, the government symbolically launched the PMJAY. The government used data from the social economic and caste census of 2011 to identify the eligible households. In the first phase, the scheme covered 107.4 million families consisting of approximately 500 million people. These are households classified as "deprived" by the above survey. I provide the details in Table A.1 in the Online Appendix. The identification is based on criteria such as the type of dwelling unit, social background, family's economic situation, etc. However, it is difficult for the authorities to verify whether a family truly meets the deprivation criteria as the data from the survey are self-reported.

Therefore, the scheme has well-defined and relatively more verifiable exclusion criteria. They are listed in Table 1. Almost all of the exclusion criteria are based on possession of some kind of assets, such as vehicles, houses and land beyond a specified size, refrigerator, etc. These types of assets usually require registration or payment of taxes. Therefore, it is relatively easier to verify their possession. Two important exclusions are worth pointing out here. First, a household gets excluded even if one of its members earns more than INR 10,000 per month. Second, a household also gets excluded if any member has an agricultural loan limit of more than INR 50,000 from a bank.

Table 1. Household Classification—Exclusions from Deprived Households

Exclusion parameters
Motorized two-/three-/four-wheeler/fishing boat
Mechanized three-/four-wheeler agricultural equipment
Kisan credit card with a credit limit of over INR 50,000
A household member is a government employee or earns more than INR 10,000 per month
Households with nonagricultural enterprises registered with the government
Any member of the household earning more than INR 10,000 per month
Paying income tax or professional tax
Paying professional tax
3 or more rooms with pucca walls and roof
Owns a refrigerator or a landline phone
Owns a landline phone
Owns more than 2.5 acres of irrigated land with 1 piece of irrigation equipment
5 acres or more of irrigated land for two or more crop season
Owning at least 7.5 acres of land or more with at least 1 piece of irrigation equipment

Notes. The table lists the exclusion criteria. The households that meet any one or more of the criteria listed are excluded from the PMJAY program.

The insurance cover is INR 500,000 per household. The insurance coverage is more than three times the average per capita income of the country when the program was launched, and hence, it is economically meaningful. Some of the other salient features of the program are as follows. First, the insurance coverage of INR 500,000 per year is for the entire family and can be utilized by one or all family members on a floater basis. There is no limit on the size of the family. Second, the scheme covers all ailments that require hospitalization, including pre-existing illnesses, at the time of the program's launch.

Third, the scheme covers both direct and incidental expenses relating to illnesses. The direct costs include all of the costs associated with treatment, including but not limited to drugs, supplies, diagnostic services, physician's fees, room charges, surgeon charges, operation theater charges, and intensive care unit charges. The incidental expenses covered include prehospitalization expenses, like the fee for laboratory and diagnostic examinations; food and accommodation expenses for the patient and one attendant; and costs relating to posthospitalization follow-up. Fourth, PMJAY has more than 20,000 empaneled hospitals, including private hospitals. Thus, by empaneling private hospitals, the government attempted to address the infrastructural challenges relating to public healthcare facilities.

Fifth, households identified as deprived were automatically enrolled in the program. The government set up kiosks outside empaneled hospitals and local government offices. Anyone could walk into these kiosks and check whether he or she is covered under the program using the biometric identity card issued by the government to all citizens. Through local authorities, the federal government distributed health cards to eligible households to ease the claim settlement process. However, the card is not necessary for eligibility.

Sixth, 60% (40%) of the program's cost is borne by the federal (state) government. Therefore, the implementation of the program requires the consent of the state governments. Finally, beneficiaries can claim insurance in any empaneled hospital anywhere in the country, not necessarily in their home state. The state governments are required to provide a list of eligible households to the federal government. Only those families included in the list by state governments are considered beneficiaries. Although most states provided such a list, some states did not. Therefore, the residents of states that did not provide the list of eligible households are not considered beneficiaries. Hence, even if they travel to other states for treatment, they cannot obtain insurance coverage under the program.

2.2.1. Total Government Spending and the Value of Benefits. As of March 31, 2020, the government issued 124.48 million health cards. More importantly, the program covered nearly 9.51 million hospital admissions in 20,257 empaneled hospitals within this period. The total government spending on reimbursement of hospital bills and incidentals as of March 31, 2020 was INR 129.45 billion. The budget allocation made by the federal and state governments up to the end of the financial year 2019–2020 was INR 146.67 billion (U.S. dollars (USD) 2.08 billion). To obtain a sense of the cost of administration, I compare the total allocation made for the program from the state and federal budgets and the amount reimbursed to the beneficiaries as insurance payments. The cost of administration so arrived at is close to 11.74% of the total budget allocation. Given that the program covers approximately 107.4 million households, the total annual cost per household is INR 1,365 (USD 19.49, 65.12 at Purchasing Power Parity) in 2019–2020. I also note that as a proportion of the overall budget and the GDP, the allocations have remained more or less constant at around 0.4% and 0.05%, respectively.

2.2.2. Program Timing. A note regarding the timing of the launch of the program is in order from an identification point of view. Although the program was launched on September 23, 2018, it took some time to set up the required public and private health infrastructure, roll out health cards, and complete the process of hospital empanelment. Therefore, not much happened in the October to December quarter of 2018, with less than 500,000 mostly pre-existing illnesses being treated. Consequently, I treat the quarter between January 2019 and March 2019 as the first quarter of the program. In January 2019 itself, the program covered more than 1.5 million hospitalizations and distributed 5 million health insurance cards.

2.2.3. The Earlier Program. It is important to note that the federal government attempted to provide health insurance to the poor through an earlier program named the "Rashtriya Swastha Bima Yojna (RSBY)." However, the program had insurance coverage of only INR 30,000 as opposed to the INR 500,000 under PMJAY. The number of hospitalizations in the first 15 months was higher than the number of hospitalizations covered by RSBY in the past 10 years. Within a year of its implementation, PMJAY added as many new empaneled hospitals as RSBY added in a decade. Thus, PMJAY can be effectively considered a new program, although it replaced RSBY.

3. Data and Descriptive Statistics

I obtained administrative data relating to loan performance from one of the credit bureaus in India. The bureau shared the microfinance loan data aggregated at a district-lender type-area type-loan cohort quarter-loan repayment quarter level. A district is a revenue unit roughly equivalent to a county in the United States. Area type refers to the classification of regions within a district as urban, semiurban, and rural. Bank type refers to the type of lender; the data set has information about microfinance loans lent by universal banks, microfinance lenders, small finance banks, cooperative banks, regional rural banks, and nonbanking finance companies.[1] Loan cohort quarter refers to the calendar quarter in which a loan was made. Finally, a loan repayment quarter refers to the calendar quarter in which the performance of the loans within an observation is assessed.

I provide sample construction details in Table 2. India is divided into 28 states and eight union territories. For ease of exposition, I call even the union territories "states." I have information about microfinance loans lent in 26 states and eight union territories. The bureau does not cover microfinance loans from the two southern states of Telangana and Andhra Pradesh.[2] Four of 36 states did not implement PMJAY. The four states are Delhi, West Bengal, Odisha, and Telangana. I call them "nonimplemented," and I call the rest "implemented." In the microfinance sample, I have data for only three nonimplemented states. In total, the data set covers 636 districts. Districts on the border of the nonimplemented states and their contiguous districts in the implemented states form the border districts sample. There are 65 such border districts belonging to 10 states—7 implemented states and 3 nonimplemented states. Figure A.1 in the Online Appendix depicts the border districts on

Table 2. Sample Construction

Sample construction	Number
Total states	34
PMJAY implemented states	31
PMJAY nonimplemented states	3
Total districts	639
PMJAY implemented districts	581
PMJAY nonimplemented districts	58
Bordering states	10
Bordering PMJAY implemented states	7
Bordering PMJAY nonimplemented states	3
Total bordering districts	65
PMJAY implemented bordering districts	36
PMJAY nonimplemented bordering districts	29
Sample period	April 2018 to March 2020
Total quarters	8
Observations of border districts sample	50,577
Observations after restricting the sample to loans lent in preimplementation	40,606
Average loan amount (INR)	32,295
Number of loans (in millions)	12
Average NPA number, %	9.83
Average NPA value, %	11.73
Standard deviation NPA number, %	24.11
Standard deviation NPA value, %	26.49

Note. The table provides sample construction details.

India's map. I have used all border districts for which the data are available. Thirty-six (29) districts belong to states that implemented (did not implement) the program.

The data set spans a period of eight quarters: from April 2018 to March 2020. Each observation (within a district, lender type, and area type) contains the quarterly loan repayment status of a bundle of loans issued in a quarter. The border districts data set has 50,577 observations relating to nearly 12 million loans. The number of observations falls to 40,606 when I restrict the data to loans made before the implementation of PMJAY. The average loan size is INR 32,295. My measures of loan performance are based on loans turning into NPAs; microfinance loans that are overdue for 90 days or more are considered NPAs. I define loan performance in two ways. The first measure, called "NPA numbers," is the proportion of loans that turn into NPAs in any quarter within an observation. The second measure, called the "NPA value," is the proportion in terms of the INR value of loans that turn into NPAs within an observation. For instance, if an observation has 10 loans with an outstanding amount of INR 100, of which the 3 loans with an outstanding amount of INR 40 are NPAs, then the NPA number (value) will be 30% (40%). The average NPA number (value) for the pre-PMJAY period is 9.83% (11.73%). The sample construction details relating to agricultural loans are in Table A.2 in the Online Appendix.

For tests relating to consumption and savings, I use the consumer pyramids data provided by the Centre for Monitoring Indian Economy. Consumer pyramids are a monthly household-level panel. The data are obtained through a survey. The data set has information about household income and various heads of expenditure, including the amount spent on loan repayment and out-of-pocket expenditures on health. The survey records whether a household borrowed a loan for medical purposes. The information about health-related loans is collected at a quarterly level. I provide details relating to the sample construction of the consumer pyramids data set in Section 1 of the Online Appendix. The data relating to the health situation come from the two rounds of the National Family Health Survey (NFHS). I consider the fourth and fifth rounds conducted in 2015–2016 and 2019–2020, respectively.

4. Empirical Strategy

For identification, I use the fact that some states did not implement the program for political reasons. The program was launched four months before the scheduled national elections. All states ruled by regional parties that were not allies of the ruling party at the federal level (Bhartiya Janata Party (BJP)) did not implement it. The states ruled by the national opposition parties—the Indian National Congress (INC) and the Communists—implemented the program. It is important to understand the likely reasons behind the divergence in stance within opposition parties to assess whether the nonimplementation by some states can be considered exogenous to the change in loan repayment behavior.

In most states, where the main national opposition party (INC) was the ruling party during the implementation of PMJAY, the BJP was the main opposition party. Therefore, in these states, the BJP had the required political base to communicate effectively to the people about its national programs. Nonimplementation of PMJAY would have given the BJP an easy poll plank in future elections. They could credibly promise to implement PMJAY if they are voted to power at both the center and state levels. It made sense for INC to implement the program to neutralize such a campaign. Further, the BJP government at the center continued all the major social sector programs launched by the previous INC-led government. Thus, despite their public bickering, the two parties seem to have some tacit understanding. Finally, given their constituency of voters, it must have been difficult for communists to stall a publicly funded health insurance program.

Regional opposition parties operate under different political circumstances. It is reasonable to assume that regional parties care more for victory in regional elections as they get an opportunity to run a state government. At the national level, they can only hope to be an ally of a leading party. Tantri and Thota (2015) show that voters vote differently in regional and national elections. Therefore, regional parties that care more about regional elections are less impacted by the BJP, making health a national election issue in 2019. Therefore, it helps regional parties to project most welfare schemes as their own. Politically, it may be profitable to give up funding from the federal government and stick to their own schemes.

In addition, in almost all states (except Delhi), where regional opposition parties dominated during the period of PMJAY implementation, the BJP did not have a significant presence, especially in regional elections. Therefore, the ability of the BJP to use PMJAY nonimplementation effectively in state elections in states dominated by regional parties was limited. This reduced the political cost of nonimplementation for regional parties.

Given these features of Indian polity, it is relatively more difficult for a national opposition party not to implement a national welfare program, such as PMJAY, than it is for a state ruled by a regional opposition party. Therefore, it is reasonable to infer that the decision to implement or not implement PMJAY was driven by exogenous political considerations.

Recognizing that states are large entities that may differ on account of several dimensions, I use a border district diff-in-diff design in line with Banerjee and Iyer (2005). I compare the bordering districts of nonimplemented states with the contiguous districts of implemented states. The difference in postimplemenation and preimplementation loan performance is the second difference. The identification assumption is that adjacent border districts of different states are likely to be similar in terms of socioeconomic characteristics, and they are also expected to be subject to similar economic shocks. In Table A.3 in the Online Appendix, I compare the border districts of nonimplemented states and the neighboring districts belonging to the implemented states in terms of population, per capita bank credit, crop yield, quantum of cultivated land, average rainfall, average loan size of microfinance and agricultural loans, health insurance coverage, and incidence of hospitalization in an average household in the pre-PMJAY period. I do not find any significant difference in any of the above variables.

An important identification requirement is that the borrowers in my sample should generally represent PMJAY beneficiaries. In this context, questions may arise regarding the use of microfinance loans. The income threshold applicable to PMJAY and microfinance is likely to drive significant overlap between PMJAY beneficiaries and the borrowers of microfinance loans. As noted in Section 2.2, a household having a member earning more than INR 120,000 a year is excluded from PMJAY. Given that the limit is at the individual level, the household income can be higher. Microfinance also has income limits as per Reserve Bank of India regulations. The applicable limits for rural areas are INR 60,000, 100,000, and 125,000 and up in financial years 2014–2015, 2018–2019, and 2019 onward, respectively. These limits are INR 100,000, 160,000, and 200,000 in financial years 2014–2015, 2018–2019, and 2019 onward, respectively, for urban areas. Further, these limits are based on household income, not individual income, as in the case of PMJAY.

Given the income limits, it is easy to see that in rural areas, there is likely to be a high overlap between PMJAY beneficiaries and microfinance borrowers for all loan-quarter cohorts. There could be a concern about the level of overlap in urban areas, especially for later cohorts. However, two points are worth noting here. Given that the PMJAY income limits are at an individual level, there could be a large number of PMJAY-eligible households in urban areas with an income of more than INR 120,000. Higher levels of expenses and relatively better employment opportunities could increase the possibility of more than one member of a family working in urban areas. Second, the median microfinance loan size in urban areas in my data is INR 30,870. Even the 90th percentile is INR 47,714. The maximum loan size permitted under microfinance is INR 75,000 in the first cycle and INR 125,000 in the second cycle. This shows that most of the microfinance borrowers in my sample are drawn from the lower strata of income, even in urban areas.

Further, given the level of household income, it is unlikely that a significant proportion of microfinance borrowers own assets, such as a large house or motorized vehicles and equipment, that lead to their disqualification. Finally, low levels of income and a lack of assets for collateral also exclude microfinance borrowers from formal banking loans. Note that the interest rate charged on microfinance loans is close to 26%, which is nearly three times the interest rate charged on bank loans. Therefore, it is likely that microfinance borrowers are those who are excluded from bank financing. Hence, it is unlikely that a significant proportion of microfinance borrowers have large bank loans that disqualify them from PMJAY.

Based on their income level, asset ownership, and access to bank finance, it is reasonable to consider microfinance borrowers representative of PMJAY beneficiaries. Nonetheless, I conduct robustness tests using farm loans where borrowers are likely to be beneficiaries.

In my primary specification, following Tantri (2018), I restrict the sample to loans issued before the implementation of PMJAY. In the absence of the above filter, loan officers' or borrowers' responses to PMJAY may mechanically lead to better loan performance in implemented regions. For instance, suppose loan officers grant a higher number of loans in implemented regions after the implementation of PMJAY. In such a case, the portfolio of loans in implemented regions mechanically will have a lower chance of turning into NPAs postimplementation because of the disproportionate presence of new loans. There will not be much time left for new loans, especially those lent toward the end of the sample period, to turn into NPAs. A similar mechanical outcome is possible if borrowers demand and obtain more loans in implemented areas. Restricting the sample to pre-existing loans ensures that I study the actual impact of PMJAY and not the anticipated effects.

5. Results—Health Insurance and Loan Performance

5.1. First-Stage Tests

I conduct a first-stage test to examine whether the program did impact health insurance coverage and out-of-pocket health expenditure. I compare the health insurance coverage in the implemented and nonimplemented regions before and after the implementation of PMJAY. I use data from both rounds of the NFHS survey. Note that the first survey was conducted before the implementation of PMJAY and that the second round was conducted during the first year of implementation. I estimate the following regression equation:

$$Y_{\mathrm{i}} = \alpha + \beta_1 \times Implemented_i + \beta_2 \times X_i + \epsilon_i. \qquad (1)$$

The data are organized at a district *i* level. I use the border district sample. Y_{i} is the change in the proportion of health insurance coverage between the two rounds of the NFHS survey in a district *i*. $Implemented_i$ is an indicator variable that takes the value of one for implemented districts and zero otherwise. X_i refers to changes in two district-level variables—rainfall and night light intensity. I cluster the errors at a district level.

I report the results in Table 3. In columns (1) and (2) in Table 3 (columns (3) and (4) in Table 3), I use the percentage point (percentage) change in health insurance coverage within a district as the dependent variable. I include the control variables in columns (2) and (4) in Table 3. As shown in Table 3, health insurance coverage increased by 15 percentage points (or 152%) more in districts belonging to the implemented states. I also conduct a diff-in-diff test with district and NFHS round fixed effects and obtain similar results in columns (5) and (6) in Table 3. Further, I find a close to 18% decline in out-of-pocket expenditure on health in implemented districts. The results are reported in Table A.4 in the Online Appendix.

5.2. Main Result—Impact on Loan Performance

I then proceed to conduct the main diff-in-diff test using microfinance loans in the border district sample by estimating the following regression equation:

$$Y_{\mathrm{itlk}} = \alpha + \beta_1 \times Implemented_i \times Post_t + \beta_2 \times X_{it} + \beta_3 \times \theta_i + \beta_4 \times \gamma_t + \beta_5 \times \omega_l + \beta_6 \times \kappa_k + \epsilon_{itlk}. \qquad (2)$$

The data are organized at a district-lender type-area type-loan cohort quarter-loan repayment quarter level. The dependent variable is a measure of loan performance. $Implemented_i$ is an indicator variable that takes the value of one for districts located in states that adopted PMJAY and zero otherwise. $Post_t$ is an

Table 3. First-Stage Test—Health Insurance Coverage

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variables	Health insurance coverage					
Implemented	14.5364***	15.6177***	141.7061**	152.2620**		
	[2.9747]	[3.0077]	[2.4308]	[2.4513]		
Implemented × *Post*					14.5364***	14.7996***
					[4.1273]	[4.3176]
log(*Rainfall*)		−1.0791		22.1117		−11.9903
		[−0.3180]		[0.5446]		[−1.0905]
log(*Night lights*)		−1.3286		−9.9965		1.5949
		[−0.3796]		[−0.2387]		[0.2143]
Observations	65	61	65	61	130	130
R^2	0.1232	0.1374	0.0857	0.1023	0.7885	0.7909
District fixed effects	No	No	No	No	Yes	Yes
Year fixed effects	No	No	No	No	Yes	Yes

Notes. The table reports the results of the first-stage test concerning health insurance coverage. In columns (1)–(4), the data are organized at a district level. I use the border district sample. In columns (1) and (2) (columns (3) and (4)), the dependent variable is the difference (ratio) between the proportion of people within a district having health insurance coverage during round 5 and round 4 of the NFHS survey. In columns (5) and (6), the data are organized at a district-round level. The dependent variable is the proportion of people within a district having health insurance coverage during the round under consideration. *Implemented* is an indicator variable that takes the value of one for districts located in states that implemented PMJAY and zero otherwise. In columns (5) and (6), *Post* is an indicator variable, taking the value of one for round 5 of the NFHS survey and zero otherwise. I include the natural logarithm of the within-districts between-rounds difference in rainfall and nightlight intensity as control variables in columns (2) and (4). In column (6), I include district-level time-varying values of the above variables as controls. I cluster the errors at the district level and adjust them for heteroscedasticity.

**5% level of significance; ***1% levels of significance.

indicator variable that takes the value of one from the second quarter of the calendar year 2019 and zero otherwise. Note that I consider the first quarter (January to April) of 2019 as the quarter of program implementation. However, it takes at least 90 days for loans to turn into NPAs. Therefore, I start assessing the postprogram performance of loans from the second calendar quarter (April to June) of 2019. I conduct robustness tests by omitting the implementation quarters.

District-level time-varying control variables included in X_{it} are the natural logarithm of rainfall in the district, the natural logarithm of the night lights, and the natural logarithm of the average loan size within an observation. The variables account for the influence of the district's weather and overall economic performance on loan performance. I also include the average loan size in INR to account for size-related effects, if any. $\theta_i(\gamma_t)(\omega_l)(\kappa_k)$ represents district (quarter)(bank-type)(area-type) fixed effects, respectively. The standard errors are clustered at a district level and adjusted for heteroscedasticity.

The results are reported in panels A and B of Table 4. The sample is restricted to loans issued before the implementation of PMJAY in panel A of Table 4. In panel B of Table 4, I consider all loans. I include district, area-type, lender-type, and quarter fixed effects in all four columns. In columns (1) and (2) in Table 4 (columns (3) and (4) in Table 4), the dependent variable is *NPA_number* (*NPA_value*). In column (2) in Table 4 (column (4) in Table 4), I include the district-level control variables described above. I find that the NPA rate of implemented districts is 3.7 (4.0) percentage points lower than the NPA rate of the nonimplemented regions in terms of *NPA_number* (*NPA_value*). Given the average sample NPA rate in terms of *NPA_number* (*NPA_value*) 9.83% (11.73%), a change of 3.4 (4.0) percentage points represents an economically meaningful 34.6% (34.1%). I find similar results in panel B of Table 4 as well. In columns (5) and (6) in Table 4, I include cohort-quarter fixed effects over and above the district-lender type-area type-loan repayment quarter-level fixed effects included in the earlier columns. The results remain largely unchanged.

5.2.1. Parallel Trends.

There could be a concern that the results presented in Table 4 merely reflect the

Table 4. Health Insurance and Loan Performance—Border District Comparison

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Loans issued before the implementation of PMJAY						
Dependent variables	*NPA_number*			*NPA_value*		
Implemented × Post	−0.0360***	−0.0337**	−0.0300**	−0.0423***	−0.0400***	−0.0354***
	[−2.7320]	[−2.6448]	[−2.4669]	[−2.9171]	[−2.9414]	[−2.7523]
Observations	32,103	30,919	30,919	31,979	30,795	30,795
R^2	0.1721	0.2719	0.6270	0.1992	0.2789	0.6445
Panel B: No restriction on loan issue date						
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variables	*NPA_number*			*NPA_value*		
Implemented × Post	−0.0301***	−0.0301***	−0.0229**	−0.0361***	−0.0361***	−0.0281***
	[−3.0077]	[−3.0077]	[−2.6014]	[−3.2017]	[−3.2017]	[−2.9024]
Observations	40,399	40,399	38,879	40,274	40,274	38,754
R^2	0.0690	0.0690	0.6342	0.0785	0.0785	0.6549
Controls	No	Yes	Yes	No	Yes	Yes
Cohort-quarter fixed effects	No	No	Yes	No	No	Yes
District fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Lender fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Area fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The table compares microfinance loans between border districts of nonimplemented states and their neighboring implemented states before and after the implementation of the PMJAY program. The data are organized at a district-lender type-area type-loan cohort quarter-loan repayment quarter level and span eight quarters from April 2018 to March 2020. In panel A, the sample is restricted to loans issued before the implementation of PMJAY. In panel B, even the loans issued after the implementation of PMJAY are considered. The dependent variable in columns (1), (2), and (3) (columns (4), (5), and (6)) is the ratio in terms of numbers (value) between loans declared as nonperforming assets and all loans within an observation. *Implemented* is a dummy variable that takes the value of one for districts located in states that adopted PMJAY and zero otherwise. *Post* is an indicator variable taking the value of one from the second quarter of 2019 and zero otherwise. I include district, quarter-level, lender-type, and area-type fixed effects in all columns and cohort-quarter fixed effects in columns (3) and (6). In columns (2), (3), (5), and (6), the district-level time-varying control variables included are the natural logarithm of the level of rainfall (measured in centimeters) and the intensity of night lights in a district quarter. I also control for average loan size. I cluster the errors at the district level and adjust them for heteroscedasticity.

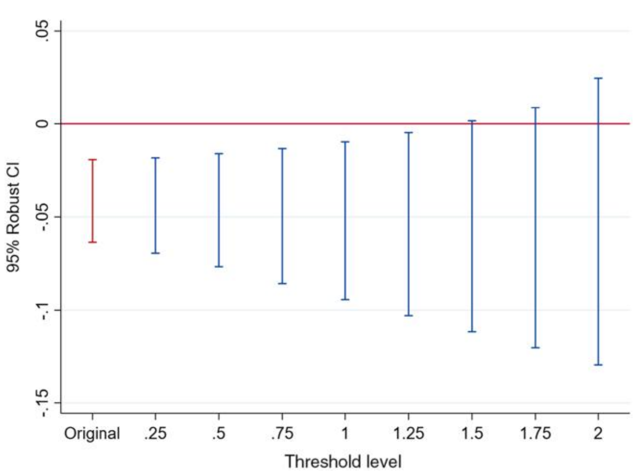
**5% level of significance; ***1% levels of significance.

continuation of a pre-existing trend. To address this concern, I use the test suggested by Rambachan and Roth (2023). I first conduct an event study with interaction terms between every quarter in the sample and the indicator variable representing the "implemented regions." The event study results are inputs to the Rambachan and Roth (2023) test. The test reveals the threshold level of pretrends required to invalidate the postevent results. A significant result at a threshold level of one means that the postevent results survive even if the continued impact of pretrends is equal to the maximum difference between the treated and control groups during the pre-event period. Therefore, a test clearing the threshold level of one should give comfort to the reader that the mechanical continuation of pretrends does not drive the postevent results. I plot the results in Figure 1, with *NPA_number* being the dependent variable. As shown in Figure 1, the results survive with threshold levels of 1 and 1.25. Even at 1.5, the *p*-value associated with the coefficient is below 6%. Given the above result, it is reasonable to conclude that the results are not because of a mechanical continuation of pretrends.

5.3. The Border District Test—Small Agricultural Loans

I conduct robustness tests using small agricultural loans. As noted in Section 2.2, borrowers of agricultural loans having a loan limit of more than INR 50,000 are excluded from PMJAY. Unfortunately, I cannot identify individual loan limits as I do not have loan-level data.

Figure 1. (Color online) Test of Pretrends Using the Rambachan and Roth (2023) Approach



Notes. In this figure, I examine the threshold level of pretrends up to which the postevent difference-in-differences results continue to be significant. I first estimate an event study with interactions between indicator variables representing each quarter in the data and the indicator variable representing the implementing regions. *NPA_number* is the dependent variable. The first quarter is in the base. I then apply the test suggested by Rambachan and Roth (2023). The horizontal axis plots the threshold levels. The vertical lines represent the 95% confidence intervals (CIs).

Therefore, I use the average loan size within an observation as the treatment variable, and I consider observations where the average loan size is INR 50,000 or less as eligible loans and those where it is more than INR 50,000 and less than INR 100,000 as ineligible loans. The identification assumption is that the proportion of beneficiaries is likely to be higher within observations having an average loan size of INR 50,000 or less.

The results are presented in Table A.5 in the Online Appendix. I first conduct a diff-in-diff test within the border districts of the implemented states by comparing the eligible and ineligible observations. The results in columns (1) and (2) in Table A.5 in the Online Appendix show that the eligible group's loan performance improves by close to 2.8 percentage points. The result depicted in Figure A.2 in the Online Appendix, which is based on the test suggested by Rambachan and Roth (2023), rules out the possibility that the mechanical continuation of pretrends drives the results in the implemented regions. I then conduct a similar diff-in-diff test using data from border districts of the nonimplemented states. As shown in columns (3) and (4) in Table A.5 in the Online Appendix, I find no significant change in the difference in loan performance between the eligible and ineligible groups. The fact that eligible loans outperform ineligible loans after the implementation of PMJAY only in implemented regions and not in similar nonimplemented regions supports the hypothesis that the implementation of PMJAY leads to an improvement in loan performance.

6. Alternative Explanations

6.1. Possible Confounding Impact of Cash Transfers

As described in Section 1, the Government of India also launched a direct cash transfer scheme exclusively for farmers. The government promised to transfer INR 6,000 per year in three installments. The first installment of INR 2,000 was transferred in April 2019 when PMJAY was being implemented. Therefore, it is essential to rule out the possibility that the results are because of the cash transfer program.

I conduct two tests to address the above concern. First, I use the fact that only one (West Bengal) of four PMJAY nonimplementing states did not implement the cash transfer. If my results hold in a subsample that excludes the state of West Bengal and its neighbors, then it is possible to rule out the impact of the cash transfer scheme. Note that in this subsample, the implemented and nonimplemented regions do not differ concerning the implementation of the cash transfer scheme. I estimate the regression Equation (2) using the above subsample. I present the results in panel A of Table 5. Notice that my results go through with similar magnitude as in Table 4.

Table 5. Health Insurance and Loan Performance—Impact of Cash Transfers

Panel A: Removing West Bengal and its neighboring states				
	(1)	(2)	(3)	(4)
Dependent variables	*NPA_number*		*NPA_value*	
Implemented × *Post*	−0.0349**	−0.0282*	−0.0453**	−0.0393**
	[−2.2110]	[−1.8730]	[−2.5015]	[−2.2996]
Observations	24,121	22,937	24,083	22,899
R^2	0.1804	0.2803	0.2086	0.2883
Panel B: Odisha vs. West Bengal (bordering districts)				
	(1)	(2)	(3)	(4)
Dependent variables	*NPA_number*		*NPA_value*	
Odisha × *Post*	0.0054	−0.0124	−0.0260	−0.0378
	[0.1618]	[−0.5405]	[−0.4566]	[−0.7038]
Observations	341	341	368	368
R^2	0.5528	0.5628	0.5172	0.5196
Control variables	No	Yes	No	Yes
District fixed effects	Yes	Yes	Yes	Yes
Year quarter fixed effects	Yes	Yes	Yes	Yes
Lender fixed effects	Yes	Yes	Yes	Yes
Area fixed effects	Yes	Yes	Yes	Yes

Notes. The table compares microfinance loans between border districts of nonimplemented and implemented states before and after the implementation of the PMJAY program, excluding West Bengal and its neighboring states in panel A. It compares eligible agricultural loans (average loan size within an observation less than or equal to INR 50,000) between the border districts of West Bengal and Odisha before and after the implementation of the PMJAY program in panel B. The data are organized at a district-lender type-area type-loan cohort quarter-loan repayment quarter level and span eight quarters from April 2018 to March 2020. The sample is restricted to loans lent before the implementation of PMJAY. The dependent variable in columns (1) and (2) (columns (3) and (4)) is the ratio in terms of numbers (value) between loans declared as nonperforming assets and all loans within an observation. *Implemented* is a dummy variable that takes the value of one for districts located in states that adopted PMJAY and zero otherwise. *Odisha* is an indicator variable that takes the value of one for districts located in Odisha and zero otherwise. *Post* is an indicator variable taking the value of one from the second quarter of 2019 and zero otherwise. I include district, quarter-level, lender-type, and area-type fixed effects in all columns. In columns (2) and (4), the district-level time-varying control variables included are the overall natural logarithm of the level of rainfall (measured in centimeters) and the intensity of night lights in a district quarter. I also control for average loan size. I cluster the errors at the district level and adjust them for heteroscedasticity.

*10% level of significance; **5% level of significance.

Second, I compare loan performance between the border districts of the neighboring states of West Bengal and Odisha. The two states do not differ concerning the implementation of PMJAY, but they differ with respect to the implementation of the cash transfer program. If my results are because of the cash transfer, loan performance is likely to be better in Odisha, which implemented the cash transfer program. Because the cash transfer is targeted at farmers, its impact should manifest clearly in agricultural loans. Therefore, I use the sample of eligible agricultural loans. I estimate the regression Equation (2) by considering Odisha as implemented and West Bengal as nonimplemented. I present the results in panel B of Table 5. I do not find a significant difference in loan performance between the two states. The results, taken together, help me rule out the cash transfer scheme as an alternative explanation.

The results also show that a cash transfer is unlikely to be a substitute for health insurance in my setting. Even with a cash transfer that is more than four times the implicit premium under PMJAY (1,350 versus 6,000), I do not see any incremental improvement in loan performance. This is not surprising for three reasons. First, the beneficiaries that I consider lack access to health insurance. Given the price and nonprice barriers described in Section 2, it is difficult for them to use the cash transfer to buy health insurance. Second, given a health shock, the insurance pays much more than the implicit premium calculated above. Thus, a cash transfer that is substantially higher than the implicit premium may not be able to mimic the payoff of an insurance policy when a household faces a health shock. Finally, cash could be used for other competing needs of the household.

6.2. Ruling Out the Ruling Party Effect

There could be a concern that the improvement in loan performance results from the favorable treatment of states where the ruling party at the federal level is also in power at the state level. Prima facie, it is unlikely that the ruling party at the federal level starts discriminating against the states ruled by the opposition parties only after the launch of PMJAY. Nonetheless, I conduct several tests to examine the above alternative explanation.

If a time-varying ruling party effect exists, then even within the implemented states, the states ruled by the federal ruling party should outperform after the PMJAY implementation. To test the above thesis, I compare the border districts of the state of Punjab with neighboring districts belonging to Himachal Pradesh and Haryana. Punjab was ruled throughout by the main opposition party—INC. The other two states were ruled out by the BJP. The ruling party effect would predict that the border districts of Haryana and Himachal Pradesh will outperform the border districts of Punjab. I estimate the regression Equation (2) and report the results in panel A of Table 6. The interaction terms are statistically indistinguishable from zero. The result is inconsistent with the ruling party effect.[3]

Similarly, the ruling party effect would predict that, within opposition-ruled states, the states that implemented PMJAY should not outperform in a diff-in-diff sense. To test the above hypothesis, I limit the sample to the opposition-ruled states of Punjab (implemented) and Delhi (nonimplemented), and I estimate the regression Equation (2). Note that although Delhi and Punjab do not share a border, they are close to each other. They are separated by the state of Haryana. I did not find any other pair of opposition-ruled states that is close to each other but differs in PMJAY implementation. I report the results in panel B of Table 6. As shown in Table 6, the NPA rate is lower by nearly three percentage points in the state of Punjab that implemented PMJAY than in the state of Delhi.

Further, it is reasonable to hypothesize that any state-wide favorable treatment leading to the outperformance of PMJAY-impacted loans should also lead to the outperformance of other types of loans where the borrowers are unlikely to be PMJAY beneficiaries. To test the above hypothesis, I estimate regression 2 separately on subsamples consisting of loans where the borrowers are unlikely to be PMJAY beneficiaries. As noted in Section 2.2, borrowers with an agricultural loan limit of greater than INR 50,000 are excluded from

Table 6. Health Insurance and Loan Performance—Time-Varying Ruling Party Effect

Panel A: Punjab vs. Haryana and Himachal Pradesh (bordering districts)				
	(1)	(2)	(3)	(4)
Dependent variables	*NPA_number*		*NPA_value*	
Punjab × Post	−0.0164 [−0.7647]	−0.0034 [−0.2086]	−0.0141 [−0.5990]	0.0000 [0.0012]
Observations	11,296	10,885	11,285	10,876
R^2	0.1637	0.2283	0.1755	0.2325
Panel B: Punjab vs. Delhi (all districts)				
	(1)	(2)	(3)	(4)
Dependent variables	*NPA_number*		*NPA_value*	
Punjab × Post	−0.0344*** [−2.8997]	−0.0302*** [−2.9230]	−0.0537*** [−3.3123]	−0.0302*** [−2.9230]
Observations	14,249	13,245	14,237	13,245
R^2	0.1738	0.1959	0.1820	0.1959
Control variables	No	Yes	No	Yes
District fixed effects	Yes	Yes	Yes	Yes
Year quarter fixed effects	Yes	Yes	Yes	Yes
Lender fixed effects	Yes	Yes	Yes	Yes
Area fixed effects	Yes	Yes	Yes	Yes

Notes. The table compares microfinance loans between border districts of the state of Punjab and its neighboring states of Himachal Pradesh and Haryana before and after the implementation of the PMJAY program in panel A. It compares microfinance loans between the states of Punjab and Delhi before and after the implementation of the PMJAY program in panel B. The data are organized at a district-lender type-area type-loan cohort-quarter level and span eight quarters from April 2018 to March 2020. The sample is restricted to loans lent before the implementation of PMJAY. The dependent variable in columns (1) and (2) (columns (3) and (4)) is the ratio in terms of numbers (value) between loans declared as nonperforming assets and all loans within an observation. *Punjab* is an indicator variable that takes the value of one for districts located in that state and zero otherwise. *Post* is an indicator variable taking the value of one from the second quarter of 2019 and zero otherwise. I include district, quarter-level, lender-type, and area-type fixed effects in all columns. In columns (2) and (4), the district-level time-varying control variables included are the overall natural logarithm of the level of rainfall (measured in centimeters) and the intensity of night lights in a district quarter. I also control for average loan size. I cluster the errors at the district level and adjust them for heteroscedasticity.

***1% levels of significance.

the program. Other exclusion criteria include ownership of a large house, ownership of motor vehicles, and an income level of more than INR 10,000 a month. However, these are required to obtain home and car loans. Banks do not offer unsecured loans and credit cards to those with an income of less than INR 10,000 a month. Given the above, I consider borrowers of large agricultural loans, home loans, vehicle loans, and credit card loans as likely excluded borrowers. I estimate the regression Equation (2) using the above category of loans. I present the results in Table A.6 in the Online Appendix. I do not find a significant change in the difference in NPA rates between the implemented and nonimplemented regions. The results are inconsistent with the ruling party effect.

6.3. Pre-Existing State-Level Insurance Schemes

A reader may worry that the nonimplemented states did not accept PMJAY because they already had health insurance schemes running and therefore, did not need it. The existing state-level health insurance schemes may make the implemented and nonimplemented states systematically different. I address this concern in several ways.

I conduct a survey of pre-existing state-level health insurance schemes. It is not true that only nonimplemented states have health insurance schemes of their own. Even implemented states, such as Rajasthan, Maharashtra, Andhra Pradesh, Gujarat, and Tamil Nadu, have their own schemes. I highlight the important features of these programs in Section 2 of the Online Appendix. None of the programs have the width and depth of PMJAY.

I conduct two specific tests in this regard. First, I compare the loan performance between the states of Telangana and Andhra Pradesh by estimating the regression Equation (2). Both of the states were a part of the united Andhra Pradesh before 2014. The erstwhile state of united Andhra Pradesh had a state-level health insurance scheme named Arogyasree. Both of the states continued the scheme after bifurcation. However, they differed with respect to PMJAY. The political party that ruled the new state of Andhra Pradesh was an ally of the federal ruling party, whereas the party that ruled Telangana was a regional opposition party. Not surprisingly, the new state of (residual) Andhra Pradesh implemented PMJAY, but the state of Telangana did not. The tests, whose results are presented in columns (1) and (2) in Table 7, show an improvement in loan performance in Andhra Pradesh. Thus, even within a sample of states with similar state-level programs, PMJAY makes a difference.[4]

Second, I conduct the above test by restricting the sample to implemented and nonimplemented states having some pre-existing health insurance of their own. The states are listed in Section 2 of the Online Appendix. The results presented in columns (3)–(6) in Table 7 show that the implemented states have a lower default rate here as well. Thus, PMJAY seems to make a difference over and above the state-level schemes.

Further, the state governments do not have the resources to implement their programs. Consequently, in many cases, the actual allocations of funds tend to be significantly restricted. For instance, as shown in

Table 7. Health Insurance and Loan Performance—Impact of Existing State-Level Schemes

	(1)	(2)	(3)	(4)	(5)	(6)
	NPA_number					
Dependent variables	Andhra–Telangana		States with local health insurance			
Implemented × *Post*	−0.0572*	−0.0660*	−0.0575***	−0.0393***	−0.0660***	−0.0796***
	[−1.8388]	[−2.0220]	[−7.8841]	[−5.7318]	[−6.6675]	[−8.1973]
Observations	1,578	1,564	155,391	131,265	55,093	52,130
R^2	0.2185	0.2280	0.1653	0.2867	0.1969	0.3292
Control variables	No	Yes	No	Yes	No	Yes
District fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Lender fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Area fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The table compares the loan performance between implemented and nonimplemented states having a local state-level health insurance scheme. In columns (1) and (2), the comparison is between the states of Andhra Pradesh (implemented) and Telangana (nonimplemented). The comparison is based on eligible agricultural loans (average size of less than or equal to INR 50,000 within an observation). The dependent variable is the proportion of loans that turn into NPAs. The comparison is based on microfinance loans in other columns. In columns (3) and (4), the comparison is between the implemented states of Assam, Chhattisgarh, Maharashtra, Gujarat, Rajasthan, Tamil Nadu, and Karnataka and the nonimplemented states of Odisha and West Bengal. All of the above states have a state-level health insurance scheme. In columns (5) and (6), I restrict the implemented states to Assam and Chhattisgarh. These states border the nonimplemented states. *Post* is an indicator variable taking the value of one from the second quarter of 2019 and zero otherwise. I include district, quarter-level, lender-type, and area-type fixed effects in all columns. In even-numbered columns, the district-level time-varying control variables included are the overall natural logarithm of the level of rainfall (measured in centimeters) and the intensity of night lights in a district quarter. I also control for average loan size. I cluster the errors at the district level and adjust them for heteroscedasticity.

*10% level of significance; ***1% levels of significance.

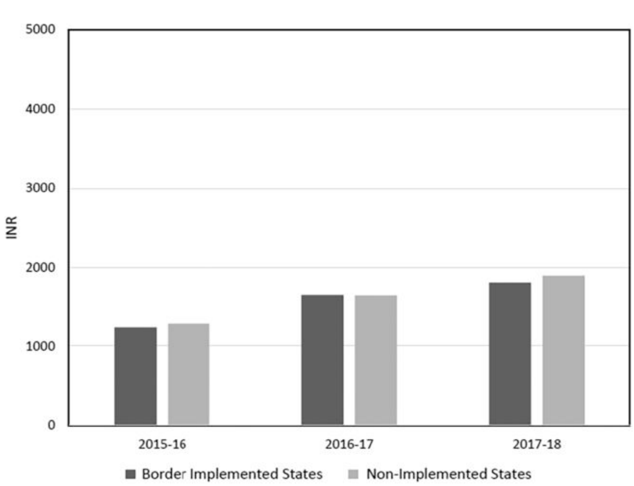
Section 2 of the Online Appendix, the state government of Odisha—one of the nonimplementing states—launched its own health insurance scheme called the Biju Swasthya Kalyan Yojna (BSKY) almost in parallel with PMJAY. The actual claims settled under BSKY are significantly lower than the health insurance claims settled under PMJAY by neighboring implementing states. The results are presented in Table A.7 in the Online Appendix. The insurance claims under BSKY are three to six times lower than claims under PMJAY. I also find that the incidence of illness/hospitalization in Odisha is no different than that in other states. Thus, the results are not because of differences in average health situation.

Finally, I test whether nonimplemented state governments allocate a higher proportion of their budget to health. The result presented in Figure 2 shows no significant difference between the nonimplemented states and their neighboring implemented states in terms of health spending. In sum, the evidence suggests that the implementation of PMJAY resulted in additional benefits even when state governments had their health insurance schemes.

6.4. Further Robustness

I address several concerns by conducting further robustness tests. Two concerns are especially noteworthy. First, a reader may worry that the results that I document may also reflect the withdrawal of RSBY described in Section 2.2.3. This is because the nonimplemented states lost access to RSBY as well. However, I think that the results that I document are because of PMJAY for two reasons. First, as described in detail in Section 2.2.3, the extent of benefits and the scale of implementation of PMJAY were an order of magnitude larger than those of RSBY. Second, as described in Section 6.3, several states, including nonimplemented states, implemented their own health insurance schemes. Even within the sample of comparable states having their own health insurance schemes, the PMJAY-implemented states outperform. As described in Section 2 of the Online Appendix, these schemes are more comprehensive than RSBY. Therefore, it is reasonable to conclude that PMJAY likely impacted loan performance even after accounting for the impact of RSBY. However, I acknowledge that I cannot fully rule out the possibility that some parts of the results that I document are because of the withdrawal of RSBY.

Figure 2. Public Health Expenditure per Capita



Notes. This figure compares the per capita public health expenditure of nonimplemented states and their neighboring states for three years before the implementation of PMJAY. I consider the expenditure from the respective state government budgets. A year here is a financial year starting on April 1 and ending on March 31.

Second, there could be a concern that the likely movement of people from the border districts of nonimplementing states to the implementing states could impact my research design. As Section 2.2 notes, the residents of nonimplemented states are not on the beneficiary list. Therefore, they cannot claim insurance under PMJAY, even if they move to other states temporarily for treatment. The residents of nonimplemented states can become beneficiaries if they migrate permanently to an implementing state. However, given the profile of the target population, permanent migration is not easy. Even if they manage to find a job and a place of residence in an implemented state, changing the identification documents and obtaining proof of residency are likely to be costly for them.

Nonetheless, I test whether PMJAY leads to significant migration from the nonimplemented states. The CMIE consumer pyramids database records whether the family surveyed has permanently migrated during the year under consideration. I estimate a regression equation similar to Equation (2) with the migration status as the dependent variable. The results reported in Table A.8 in the Online Appendix show no significant difference between the implemented and nonimplemented states in terms of migration.

The other robustness tests that I perform are the following. (i) In Table A.9 in the Online Appendix, I compare all districts in nonimplemented states with all districts of implemented states and find results consistent with my main results. (ii) I leave out the two implementation quarters (the fourth quarter of 2018 and the first quarter of 2019) to leave some time for hospitals' empanelment and setting up other facilities. I present the results in Table A.10 in the Online Appendix. My results remain largely unchanged. (iii) To account for any announcement effect, I further restrict the sample to loans lent at least two years before the implementation of the program and estimate the regression Equation (2). The results presented in Table A.11 in the Online Appendix are of similar magnitude as my main results. (iv) To rule out any kind of mechanical alternative explanation, in Tables A.12, A.13, and A.14 in the

Online Appendix, I present the results of several placebo tests. The placebo tests show no significant difference in loan performance between the implemented and nonimplemented states. (v) To address any residual concerns about the representativeness of my sample, I estimate the regression Equation (2) by limiting the sample to agricultural loan cohorts having an average loan value of less than INR 50,000. As described in Section 2.2, borrowers of smaller loans and rural borrowers are more likely to be representative PMJAY beneficiaries. The results are presented in Table A.15 in the Online Appendix. The threshold considered in columns (1) and (2) in Table A.15 in the Online Appendix is INR 40,000. The threshold considered in columns (3) and (4) in Table A.15 in the Online Appendix is INR 30,000. The threshold considered in columns (5) and (6) in Table A.15 in the Online Appendix is INR 20,000. In columns (7) and (8) in Table A.15 in the Online Appendix, I limit the sample to microfinance loans issued in rural areas. As shown in Table A.15 in the Online Appendix, the results are in line with my main results.

7. The Mechanism

7.1. Improvement in the Liquidity Available for Loan Repayment

The liquidity channel could work in the following three ways. First, it is conceivable that poor households accumulate precautionary savings to deal with emergencies, such as health shocks. Such households may prioritize accumulating such savings over loan repayment. Thus, building precautionary savings could lead to loan defaults when households face shocks to their income. The PMJAY program reduces the need to maintain precautionary savings from a health expenditure point of view. Thus, it increases the liquidity available for loan repayment.

To investigate the above channel, I calculate precautionary savings as the difference between income and expenditure using the CMIE data. Expenditure includes the amount spent on loan repayments and not the amount payable. Thus, a household can increase savings by defaulting on loans. I take a comprehensive view of savings to include savings in both financial and physical assets. I recognize that a significant portion of poor households face difficulty in accessing formal financial savings instruments and that they may use investments in physical assets, such as gold, for savings (Gertler and Gruber 2002).

I consider PMJAY-eligible households within the border districts of implemented and nonimplemented districts. I use data on income and assets to identify plausible beneficiaries. Because the need for precautionary savings is likely to be prevalent among the poorest households, I consider the bottom 10% of the households in terms of their monthly income. I then estimate a diff-in-diff regression equation similar to Equation (2).

The results are presented in Table 8. In column (1) in Table 8, the dependent variable is the ratio between precautionary savings and income at a household-month level. I find a more than 1.5 times decline in the precautionary savings as a proportion of income. In column (2) in Table 8, I further restrict the data to

Table 8. Health Insurance and Loan Performance—The Liquidity Channel

	Bottom 10 percentile families who are also eligible for PMJAY			Upper 10 percentile families		
	Savings/income		Borrowed for medical expenses	Savings/income		Borrowed for medical expenses
Dependent variables	All families	Healthy families	All families	All families	Healthy families	All families
Implemented × *Post*	−1.4226*	−2.3377*	−0.0579**	0.0241**	0.0223*	−0.0012
	[−1.7001]	[−1.9989]	[−2.0613]	[2.1600]	[1.9547]	[−0.3890]
Observations	19,537	9,051	7,235	52,596	30,877	15,263
R^2	0.0750	0.0691	0.0883	0.1039	0.1101	0.0124
District fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year month fixed effects	Yes	Yes	No	Yes	Yes	No
Year wave fixed effects	No	No	Yes	No	No	Yes

Notes. The data are organized at a household-month (household-wave) level and restricted to border districts of nonimplementing states and their neighboring implementing states in columns (1), (2), (4), and (5) (columns (3) and (6)). The dependent variable is the ratio between savings and income in columns (1), (2), (4), and (5). In columns (3) and (6), the dependent variable is an indicator variable that takes the value of one if the household under consideration borrows a loan for health purposes during the survey wave under consideration and zero otherwise. *Implemented* is an indicator variable that takes the value of one for districts located in states that adopted PMJAY and zero otherwise. *Post* is an indicator variable taking the value of one for postimplementation months and zero otherwise. I include district and year-month (year-wave) fixed effects in columns (1), (2), (4), and (5) (columns (3) and (6)). A survey wave is a period of four months starting from January. The survey wave ending in April 2019 is considered the first survey wave after implementation. The postperiod starts from that wave for the purpose of columns (3) and (6). The data are restricted to the PMJAY beneficiary households that are in the bottom 10% in terms of income in columns (1), (2), and (3). In columns (4), (5), and (6), the data are restricted to households that are a part of the top 10% in terms of income. The errors are clustered at a district level and adjusted for heteroskedasticity.

*10% level of significance; **5% level of significance.

households that report facing no health-related issues within the family during the survey period. I find a decline of more than two times. In column (1) in Table A.16 in the Online Appendix, I rule out the existence of pretrends in precautionary savings between the implemented and nonimplemented regions. Therefore, it is reasonable to conclude that the result represents a reduction in precautionary savings. The fact that households not facing health shocks reduce savings further supports the precautionary savings hypothesis. To the extent that precautionary savings are prioritized over loan repayments, the reduction in precautionary savings represents additional liquidity available for loan repayment.

It is possible that a decision to default on a loan for building precautionary savings in the pre-PMJAY period involved a consideration of the likely haircut and the future consequences of default. Given that contract enforcement is weak in India, it is reasonable to hypothesize that a borrower gets to keep almost the entire amount in default (Mukherjee et al. 2018). This does not mean that defaulting on a loan does not have a cost. It is well established that in emerging economies with weak contract enforcement mechanisms, lenders use the threat of denial of future loans to enforce loan repayment (Bond and Rai 2009, Schiantarelli et al. 2020). In the case of microfinance loans, the likely loss of social capital is the dominant enforcement mechanism (Feigenberg et al. 2013). This implies that borrowers are likely to default only when they think that the utility of having precautionary savings for health purposes is higher than the loss because of losing access to credit in the future. My results suggest that having publicly funded health insurance may reduce the need to choose between maintaining the requisite savings for health and the consequences of a loan default.

Next, I examine whether the households borrowed less for health purposes after the implementation of PMJAY. If PMJAY worked as intended, then it is reasonable to expect a reduction in such borrowing. The CMIE collects information about health-related borrowing once every four months. They call the four-month interval within which this information is collected as a survey wave. Therefore, to examine the impact of PMJAY on health-related loans, I organize the data at a household-survey-wave level. The survey wave ending in April 2019 is considered the first survey after PMJAY implementation.

I test the above thesis by estimating regression Equation (2). The survey only has data about the existence of a health-related loan and does not have data about the loan amount. Therefore, the dependent variable is an indicator variable that represents the existence of a health-related loan. I present the results in column (3) in Table 8. As expected, I find a 5.79-percentage-point decline in borrowings for health purposes. The second column in Table A.16 shows that the decline in health-related borrowing is not a result of a continuation of a pre-existing trend.

In Table A.17 in the Online Appendix, I examine the sources of borrowing for health emergencies. I find a significant decline in borrowings from money lenders, who usually lend at exorbitant rates. I also find a decline in borrowings from family and friends. However, the coefficient estimate is not statistically significant at conventional levels. I do not find a significant change in asset sales.

Finally, as discussed in Section 5.1, the results presented in Table A.4 in the Online Appendix show a decline in out-of-pocket expenditure on health. The same result holds even when I restrict the sample to low-income households.

For robustness, I conduct placebo tests by restricting the sample to the top 10% of the households in terms of income. These households are unlikely to be PMJAY beneficiaries. In the results presented in columns (4), (5), and (6) in Table 8, I do not detect any significant reduction in savings or any change in the tendency to borrow for health purposes.

In sum, the results suggest that beneficiary households reduce financial savings and expensive informal borrowings in response to the PMJAY implementation. They also experience a reduction in out-of-pocket expenditure on health. Thus, liquidity available for loan repayments gets considerably enhanced.

7.2. The Health Channel

The program may improve overall health in the long run by increasing access to preventive and curative care. This could enhance productivity, increase income, and consequently, lead to better loan performance. Because I consider a short period of only four quarters after the intervention, I cannot provide credible evidence on this front.

8. Discussion

8.1. Overall Costs and Benefits

To get a full picture of the impact of PMJAY, I evaluate the overall expenditure, the value of the implicit benefit received by beneficiaries, and the possibility of delivering the same value through alternative policy designs.

As noted in Section 2.2.1, the overall budget allocation on PMJAY approximates INR 1,365 per household at the end of the financial year 2020. There are two other plausibly independent ways of calculating the benefits. A change in the difference in the out-of-pocket expenditure on healthcare between the households in the implemented and nonimplemented regions in response to PMJAY gives a reasonable estimate of the implicit savings because of PMJAY. The results presented in Table A.4 in the Online Appendix show a reduction in out-of-pocket expenses of close to 17.2%. Given the pre-PMJAY average spending of INR 4,400 per household, the decline for treated households is nearly INR 760 per year. The third way is by estimating the NPAs saved. The overall NPA reduction is nearly INR 96 billion (4% of the microfinance loan book of 2.4 trillion) or 960 per household in terms of the reduction in interest burden. Implicit is the assumption that the recovery rate from NPAs is zero. This is a reasonable assumption when contract enforcement is inefficient. If the credit markets are not competitive, a part of the gain will go to the lender.

It is important to explain the differences between the per-household spending by the government and the implicit benefits. First, it is possible that some households had to forgo treatment before PMJAY because of a lack of funds. The out-of-pocket health expenditure before the program and consequently, the benefits under the program are understated in such cases. For instance, consider a poor household forgoing treatment worth INR 600,000 before PMJAY. Suppose that after PMJAY, they avail the treatment, and they claim the insurance coverage of INR 500,000 and spend INR 100,000. In this case, I will significantly understate the benefits by considering the pre-PMJAY spending as zero. Second, administrative costs account for some of the differences. As explained in Section 2.2.1, the administrative costs are close to 11%. Finally, given that I do not perfectly identify beneficiaries in the treated group, there could be a systematic underestimation of the treatment effect. Therefore, it is reasonable to claim that the unconditional implicit benefit because of the program is between INR 760 to INR 1,300 per household per year.

Finally, I compare the magnitude of benefits under PMJAY with the amount required for loan repayment. The implicit insurance premium incurred by the government is around INR 1,300 per household. The average claim amount in private hospitals is nearly INR 17,000. The average borrowing household spends around INR 15,000 on loan repayment for a three-year loan. Thus, the expense incurred per household, given a health shock, is close to the amount required for loan repayment. The numbers explain why a family may have to choose between loan repayment and saving for health emergencies when they face income shocks. By spending a fraction of the amount required to be spent by households, the government obviates the need for large savings.

8.2. Relevance Beyond India

Given several unique aspects of the program and the institutional details, a reader may wonder whether these findings generalize beyond the specific contours of the Indian setting. To understand the generalizability of my findings to other similar countries, I looked at the Findex database (2021) maintained by the World Bank and the Comparative Health Policy Library maintained by the Irving Medical Center of Columbia University. The former database provides information about the situation pertaining to access to credit and the importance of healthcare expenditure in countries, whereas the latter provides information about the major healthcare schemes either being implemented or being planned in several countries.

In Table A.18 in the Online Appendix, I compare India with 14 other comparable countries in terms of credit market development and the importance of medical costs. The countries that I consider are the other four countries in the group of Brazil, Russia, India, China, and South Africa; other comparable countries such as Indonesia and Argentina; and some of the neighboring countries. As shown in Table A.18 in the Online Appendix, India stands closer to the median among these countries in terms of the proportion of the population having bank accounts, the proportion of the population that has borrowed in the last year, and the degree to which expenditure on healthcare is the most worrying financial issue facing the households.

From the Columbia University database, we learn that many of these countries face conditions similar to India's conditions in terms of health insurance and that they have either launched or are planning to launch a publicly funded health insurance program similar to PMJAY. For instance, the Sistema Único de Saúde (SUS) is Brazil's national health system that aims to achieve universal health coverage. As in the case of PMJAY, the program is administered by both the federal government and the state governments. Further, like PMJAY, SUS offers many services free of charge, and it covers both preventive care and actual out-of-pocket health expenditures. The health insurance situation in Mexico is even closer to that of India. Out-of-pocket health expenditure is close to 45% of the total expenditure on health. The country started a new publicly funded health insurance scheme administered by the Institute of Health for Wellbeing in January 2020. The program aims to cover the segment of the population not covered by either employer-supported or private health insurance schemes. Thus, the program is comparable with PMJAY. However, the discussion in the media suggests that the program is facing several implementation-related hurdles. The Universal Coverage Scheme of Thailand is also publicly funded, and it covers the segment of the population not having access to private health insurance. South Africa is planning to launch a PMJAY-like universal health insurance program by 2026. Indonesia also has similar plans.

However, even after an extensive search, I did not find any systematic study about the spillover effects of these programs on credit markets. My findings are likely to be relevant for assessing the likely impact of existing such programs in other similar countries and also, for countries that are planning to launch a universal healthcare program. However, my findings are not generalizable to settings where access to formal credit and market-driven health insurance is easily available.

9. Conclusion

I study the impact of the world's largest publicly funded health insurance plan on loan repayment behavior. PMJAY launched by the Government of India covers nearly 500 million people. The insurance plan is comprehensive, and it covers even pre-existing illnesses and incidental expenses related to hospitalization.

For identification, I use the fact that states ruled by regional opposition parties did not implement the program for political reasons. Comparing border regions of states that did not implement the program with contiguous areas belonging to states that implemented the program within a diff-in-diff framework, I find that the program's introduction is associated with a significant improvement in loan delinquency. An examination of the data from a national household-level income and consumption survey shows that the improvement in loan performance is most likely because of an increase in liquidity for loan repayment. Liquidity improves because of a reduction in precautionary savings, out-of-pocket expenditure, and indebtedness relating to health.

A limitation of this study is that I do not examine the impact of PMJAY on access to credit in general. Because I lack data on loan applications, it is hard to disentangle the demand-side effects from the supply-side effects. Also, the program may have different types of effects based on the nature of credit. For instance, it may reduce the demand for emergency credit and increase the demand for credit, leading to long-term productivity. Also, I do not examine the long-run consequences in this study.

Endnotes

[1] Small finance banks are banks that specialize in small-ticket loans. Cooperative banks are community-owned banks. Regional rural banks are banks that are mandated to serve rural areas.

[2] The erstwhile state of Andhra Pradesh, from which the two states were carved in 2014, suffered a major microfinance crisis in the year 2010 because of an unexpected government intervention that made loan recovery almost impossible. Since then, the bureau does not collect microfinance loan data from the two states.

[3] Note that I cannot conduct such a border district test with other opposition-ruled states. Kerela was ruled by the Communist Party of India (Marxist). During the sample period, none of the neighboring states of Kerala were ruled by the BJP. BJP came to power in neighboring Karnataka during the year 2019. Chattisgarh, Rajasthan, and Madhya Pradesh were ruled by the BJP until late 2018. The INC came to power in December 2018 in these states. It is likely that the previous BJP governments had a role in the implementation of PMJAY in these states.

[4] Note that I can only conduct the above test using the sample of eligible agricultural loans as data relating to microfinance loans are unavailable for Telangana and Andhra Pradesh.

References

Abaluck J, Bravo MMC, Hull P, Starc A (2021) Mortality effects and choice across private health insurance plans. *Quart. J. Econom.* 136(3):1557–1610.

Agarwal S, Mazumder B (2013) Cognitive abilities and household financial decision making. *Amer. Econom. J. Appl. Econom.* 5(1):193–207.

Allen H, Swanson A, Wang J, Gross T (2017) Early Medicaid expansion associated with reduced payday borrowing in California. *Health Affairs* 36(10):1769–1776.

Anagol S, Cole S, Sarkar S (2017) Understanding the advice of commissions-motivated agents: Evidence from the Indian life insurance market. *Rev. Econom. Statist.* 99(1):1–15.

Badarinza C, Balasubramaniam V, Ramadorai T (2019) The household finance landscape in emerging economies. *Annual Rev. Financial Econom.* 11:109–129.

Badarinza C, Campbell JY, Ramadorai T (2016) International comparative household finance. *Annual Rev. Econom.* 8:111–144.

Baicker K, Finkelstein A, Song J, Taubman S (2014) The impact of Medicaid on labor market activity and program participation: Evidence from the Oregon health insurance experiment. *Amer. Econom. Rev.* 104(5):322–328.

Banerjee A, Iyer L (2005) History, institutions, and economic performance: The legacy of colonial land tenure systems in India. *Amer. Econom. Rev.* 95(4):1190–1213.

Beck T, Webb I (2003) Economic, demographic, and institutional determinants of life insurance consumption across countries. *World Bank Econom. Rev.* 17(1):51–88.

Bharadwaj P, Løken KV, Neilson C (2013) Early life health interventions and academic achievement. *Amer. Econom. Rev.* 103(5): 1862–1891.

Bond P, Rai AS (2009) Borrower runs. *J. Development Econom.* 88(2):185–191.

Coile CC (2004) Health shocks and couples' labor supply decisions. NBER Working Paper No. 10810, National Bureau of Economic Research, Cambridge, MA.

Cole SA, Xiong W (2017) Agricultural insurance and economic development. *Annual Rev. Econom.* 9:235–262.

Cole S, Giné X, Vickery J (2017) How does risk management influence production decisions? Evidence from a field experiment. *Rev. Financial Stud.* 30(6):1935–1970.

Cole S, Giné X, Tobacman J, Topalova P, Townsend R, Vickery J (2013) Barriers to household risk management: Evidence from India. *Amer. Econom. J. Appl. Econom.* 5(1):104–135.

Das J, Holla A, Mohpal A, Muralidharan K (2016) Quality and accountability in healthcare delivery: Audit-study evidence from primary care in India. *Amer. Econom. Rev.* 106(12): 3765–3799.

Dobkin C, Finkelstein A, Kluender R, Notowidigdo MJ (2018) The economic consequences of hospital admissions. *Amer. Econom. Rev.* 108(2):308–352.

Dupas P, Keats A, Robinson J (2019) The effect of savings accounts on interpersonal financial relationships: Evidence from a field experiment in rural Kenya. *Econom. J.* 129(617):273–310.

Dupas P, Karlan D, Robinson J, Ubfal D (2018) Banking the unbanked? Evidence from three countries. *Amer. Econom. J. Appl. Econom.* 10(2):257–297.

Feigenberg B, Field E, Pande R (2013) The economic returns to social interaction: Experimental evidence from microfinance. *Rev. Econom. Stud.* 80(4):1459–1483.

Fernandes D, Lynch JG Jr, Netemeyer RG (2014) Financial literacy, financial education, and downstream financial behaviors. *Management Sci.* 60(8):1861–1883.

Finkelstein A, McKnight R (2008) What did Medicare do? The initial impact of Medicare on mortality and out of pocket medical spending. *J. Public Econom.* 92(7):1644–1668.

Finkelstein A, Taubman S, Wright B, Bernstein M, Gruber J, Newhouse JP, Allen H, Baicker K; Oregon Health Study Group

(2012) The Oregon health insurance experiment: Evidence from the first year. *Quart. J. Econom.* 127(3):1057–1106.

Flory JA (2018) Formal finance and informal safety nets of the poor: Evidence from a savings field experiment. *J. Development Econom.* 135:517–533.

Gallagher EA, Gopalan R, Grinstein-Weiss M (2019) The effect of health insurance on home payment delinquency: Evidence from ACA marketplace subsidies. *J. Public Econom.* 172:67–83.

Garthwaite C, Gross T, Notowidigdo MJ (2014) Public health insurance, labor supply, and employment lock. *Quart. J. Econom.* 129(2):653–696.

Gaurav S, Cole S, Tobacman J (2011) Marketing complex financial products in emerging markets: Evidence from rainfall insurance in India. *J. Marketing Res.* 48(Suppl):S150–S162.

Gertler P, Gruber J (2002) Insuring consumption against illness. *Amer. Econom. Rev.* 92(1):51–70.

Goldin J, Lurie IZ, McCubbin J (2021) Health insurance and mortality: Experimental evidence from taxpayer outreach. *Quart. J. Econom.* 136(1):1–49.

Goldsmith-Pinkham PS, Pinkovskiy M, Wallace J (2020) Medicare and the geography of financial health. FRB of New York Staff Report No. 911, Federal Reserve Bank of New York, New York.

Gomes F, Haliassos M, Ramadorai T (2021) Household finance. *J. Econom. Literature* 59(3):919–1000.

Goodman-Bacon A (2018) Public insurance and mortality: Evidence from Medicaid implementation. *J. Political Econom.* 126(1):216–262.

Gruber J, Simon K (2008) Crowd-out 10 years later: Have recent public insurance expansions crowded out private health insurance? *J. Health Econom.* 27(2):201–217.

Hu L, Kaestner R, Mazumder B, Miller S, Wong A (2018) The effect of the Affordable Care Act Medicaid expansions on financial well-being. *J. Public Econom.* 163:99–112.

Karlan D, McConnell M, Mullainathan S, Zinman J (2016) Getting to the top of mind: How reminders increase saving. *Management Sci.* 62(12):3393–3411.

Levine D, Polimeni R, Ramage I (2016) Insuring health or insuring wealth? An experimental evaluation of health insurance in rural Cambodia. *J. Development Econom.* 119:1–15.

Mahoney N (2015) Bankruptcy as implicit health insurance. *Amer. Econom. Rev.* 105(2):710–746.

Miller S, Johnson N, Wherry LR (2021a) Medicaid and mortality: New evidence from linked survey and administrative data. *Quart. J. Econom.* 136(3):1783–1829.

Miller S, Hu L, Kaestner R, Mazumder B, Wong A (2021b) The ACA Medicaid expansion in Michigan and financial health. *J. Policy Anal. Management* 40(2):348–375.

Mukherjee S, Subramanian K, Tantri P (2018) Borrowers' distress and debt relief: Evidence from a natural experiment. *J. Law Econom.* 61(4):607–635.

Rambachan A, Roth J (2023) A more credible approach to parallel trends. *Rev. Econom. Stud.* 90(5):2555–2591.

Schiantarelli F, Stacchini M, Strahan PE (2020) Bank quality, judicial efficiency, and loan repayment delays in Italy. *J. Finance* 75(4):2139–2178.

Shigeoka H (2014) The effect of patient cost sharing on utilization, health, and risk protection. *Amer. Econom. Rev.* 104(7):2152–2184.

Tantri PL (2018) Contagious effects of a political intervention in debt contracts: Evidence using loan-level data. *Rev. Financial Stud.* 31(11):4556–4592.

Tantri PL, Thota N (2015) Effective democracy and swing voters: Evidence based on actual voting records. Indian School of Business research paper, Indian School of Business, Hyderabad, India.

Wagstaff A (2007) The economic consequences of health shocks: Evidence from Vietnam. *J. Health Econom.* 26(1):82–100.